

Franka Emika Robot Control Comparison Report

Automatic Analysis

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1 Overview

This report presents a comprehensive comparison of robot control performance across different:

- Controllers: FBL (Feedback Linearization) and PBC (Passivity-Based Control)
- Uncertainty levels: high, extreme
- Initial conditions: matched vs mismatched
- Gain tuning: optimized vs unoptimized
- Trajectory types: quintic, bang-bang, bang-coast-bang

Total simulations analyzed: 96

2 Controller Performance Summary

2.1 FBL Controller

Metric	Value
Number of simulations	48
Mean EE max error [mm]	31.54
Std EE max error [mm]	23.30
Best EE max error [mm]	3.88
Worst EE max error [mm]	83.23
Mean joint max error [deg]	11.85
Best joint max error [deg]	0.60
Worst joint max error [deg]	20.59

Table 1: Summary statistics for FBL controller

2.2 PBC Controller

Metric	Value
Number of simulations	48
Mean EE max error [mm]	13.41
Std EE max error [mm]	8.01
Best EE max error [mm]	1.41
Worst EE max error [mm]	20.84
Mean joint max error [deg]	5.65
Best joint max error [deg]	0.34
Worst joint max error [deg]	10.00

Table 2: Summary statistics for PBC controller

3 Detailed Performance Tables

Performance tables show results grouped by trajectory type, with test labels A and B, including averages for each configuration. Lower values indicate better performance.

3.1 Uncertainty Level: HIGH

3.1.1 Matched Initial Conditions, Optimized Gains

Table 3: FBL Controller - selective uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0059	0.34	3.20
	B	0.0076	0.43	8.67
Bang-Bang	A	0.0059	0.34	3.58
	B	0.0076	0.43	10.69
Bang-Coast-Bang	A	0.0058	0.33	3.63
	B	0.0074	0.42	10.36
Average		0.0067	0.38	6.69

Table 4: PBC Controller - selective uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0031	0.18	2.83
	B	0.0071	0.41	3.90
Bang-Bang	A	0.0030	0.17	2.93
	B	0.0068	0.39	3.87
Bang-Coast-Bang	A	0.0030	0.17	2.95
	B	0.0067	0.39	3.88
Average		0.0050	0.28	3.39

3.1.2 Matched Initial Conditions, Unoptimized Gains

Table 5: FBL Controller - selective uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0933	5.35	12.06
	B	0.0898	5.15	35.40
Bang-Bang	A	0.0925	5.30	17.10
	B	0.0891	5.10	32.77
Bang-Coast-Bang	A	0.0929	5.32	16.98
	B	0.0885	5.07	32.75
Average		0.0910	5.22	24.51

Table 6: PBC Controller - selective uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0078	0.45	6.78
	B	0.0099	0.57	6.63
Bang-Bang	A	0.0076	0.44	7.05
	B	0.0099	0.57	6.89
Bang-Coast-Bang	A	0.0076	0.44	7.08
	B	0.0099	0.57	6.89
Average		0.0088	0.50	6.89

3.1.3 Mismatched Initial Conditions, Optimized Gains

Table 7: FBL Controller - selective uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0245	1.40	3.20
	B	0.0215	1.23	8.67
Bang-Bang	A	0.0255	1.46	3.58
	B	0.0214	1.23	10.69
Bang-Coast-Bang	A	0.0256	1.46	3.63
	B	0.0217	1.24	10.36
Average		0.0234	1.34	6.69

Table 8: PBC Controller - selective uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0309	1.77	2.83
	B	0.0337	1.93	3.90
Bang-Bang	A	0.0346	1.98	2.94
	B	0.0347	1.99	3.87
Bang-Coast-Bang	A	0.0341	1.95	2.94
	B	0.0347	1.99	3.88
Average		0.0338	1.94	3.39

3.1.4 Mismatched Initial Conditions, Unoptimized Gains

Table 9: FBL Controller - selective uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0932	5.34	15.29
	B	0.0908	5.20	35.39
Bang-Bang	A	0.0924	5.29	16.91
	B	0.0901	5.16	32.77
Bang-Coast-Bang	A	0.0927	5.31	16.70
	B	0.0901	5.16	32.75
Average		0.0915	5.24	24.97

Table 10: PBC Controller - selective uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0320	1.84	6.78
	B	0.0267	1.53	6.63
Bang-Bang	A	0.0351	2.01	7.05
	B	0.0271	1.55	6.89
Bang-Coast-Bang	A	0.0350	2.00	7.08
	B	0.0259	1.48	6.90
Average		0.0303	1.74	6.89

3.2 Uncertainty Level: EXTREME

3.2.1 Matched Initial Conditions, Optimized Gains

Table 11: FBL Controller - uniform uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0060	0.34	2.88
	B	0.0095	0.55	18.57
Bang-Bang	A	0.0058	0.33	2.72
	B	0.0093	0.54	20.56
Bang-Coast-Bang	A	0.0058	0.33	2.77
	B	0.0091	0.52	20.53
Average		0.0076	0.44	11.34

Table 12: PBC Controller - uniform uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0032	0.18	1.32
	B	0.0088	0.50	5.48
Bang-Bang	A	0.0030	0.17	1.38
	B	0.0085	0.49	5.44
Bang-Coast-Bang	A	0.0030	0.17	1.38
	B	0.0084	0.48	5.45
Average		0.0058	0.33	3.41

3.2.2 Matched Initial Conditions, Unoptimized Gains

Table 13: FBL Controller - uniform uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.1187	6.80	30.65
	B	0.1168	6.69	80.51
Bang-Bang	A	0.1186	6.79	30.99
	B	0.1161	6.65	82.66
Bang-Coast-Bang	A	0.1176	6.74	30.86
	B	0.1143	6.55	81.74
Average		0.1170	6.70	56.23

Table 14: PBC Controller - uniform uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0086	0.49	2.76
	B	0.0126	0.72	10.12
Bang-Bang	A	0.0084	0.48	2.84
	B	0.0125	0.72	10.38
Bang-Coast-Bang	A	0.0084	0.48	2.84
	B	0.0124	0.71	10.42
Average		0.0105	0.60	6.56

3.2.3 Mismatched Initial Conditions, Optimized Gains

Table 15: FBL Controller - uniform uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0265	1.52	2.88
	B	0.0227	1.30	18.60
Bang-Bang	A	0.0268	1.54	2.72
	B	0.0231	1.32	20.57
Bang-Coast-Bang	A	0.0267	1.53	2.77
	B	0.0230	1.32	20.53
Average		0.0248	1.42	11.34

Table 16: PBC Controller - uniform uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0314	1.80	1.32
	B	0.0340	1.95	5.48
Bang-Bang	A	0.0347	1.99	1.38
	B	0.0351	2.01	5.44
Bang-Coast-Bang	A	0.0343	1.97	1.38
	B	0.0353	2.02	5.45
Average		0.0341	1.96	3.41

3.2.4 Mismatched Initial Conditions, Unoptimized Gains

Table 17: FBL Controller - uniform uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.1184	6.78	34.02
	B	0.1157	6.63	64.76
Bang-Bang	A	0.1180	6.76	34.41
	B	0.1153	6.61	66.10
Bang-Coast-Bang	A	0.1170	6.71	34.27
	B	0.1143	6.55	66.40
Average		0.1164	6.67	49.99

Table 18: PBC Controller - uniform uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0327	1.87	2.76
	B	0.0292	1.67	10.12
Bang-Bang	A	0.0355	2.03	2.84
	B	0.0304	1.74	10.38
Bang-Coast-Bang	A	0.0354	2.03	2.84
	B	0.0304	1.74	10.42
Average		0.0323	1.85	6.56

4 FBL vs PBC Controller Comparison

This section provides direct head-to-head comparisons between FBL and PBC controllers under identical conditions.

4.1 Uncertainty Level: HIGH

4.1.1 Matched Initial Conditions, Optimized Gains

Table 19: FBL vs PBC Comparison - selective uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0059	0.34	3.20
	PBC	A	0.0031	0.18	2.83
	FBL	B	0.0076	0.43	8.67
	PBC	B	0.0071	0.41	3.90
Bang-Bang	FBL	A	0.0059	0.34	3.58
	PBC	A	0.0030	0.17	2.93
	FBL	B	0.0076	0.43	10.69
	PBC	B	0.0068	0.39	3.87
Bang-Coast-Bang	FBL	A	0.0058	0.33	3.63
	PBC	A	0.0030	0.17	2.95
	FBL	B	0.0074	0.42	10.36
	PBC	B	0.0067	0.39	3.88

4.1.2 Matched Initial Conditions, Unoptimized Gains

Table 20: FBL vs PBC Comparison - selective uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0933	5.35	12.06
	PBC	A	0.0078	0.45	6.78
	FBL	B	0.0898	5.15	35.40
	PBC	B	0.0099	0.57	6.63
Bang-Bang	FBL	A	0.0925	5.30	17.10
	PBC	A	0.0076	0.44	7.05
	FBL	B	0.0891	5.10	32.77
	PBC	B	0.0099	0.57	6.89
Bang-Coast-Bang	FBL	A	0.0929	5.32	16.98
	PBC	A	0.0076	0.44	7.08
	FBL	B	0.0885	5.07	32.75
	PBC	B	0.0099	0.57	6.89

4.1.3 Mismatched Initial Conditions, Optimized Gains

Table 21: FBL vs PBC Comparison - selective uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0245	1.40	3.20
	PBC	A	0.0309	1.77	2.83
	FBL	B	0.0215	1.23	8.67
	PBC	B	0.0337	1.93	3.90
Bang-Bang	FBL	A	0.0255	1.46	3.58
	PBC	A	0.0346	1.98	2.94
	FBL	B	0.0214	1.23	10.69
	PBC	B	0.0347	1.99	3.87
Bang-Coast-Bang	FBL	A	0.0256	1.46	3.63
	PBC	A	0.0341	1.95	2.94
	FBL	B	0.0217	1.24	10.36
	PBC	B	0.0347	1.99	3.88

4.1.4 Mismatched Initial Conditions, Unoptimized Gains

Table 22: FBL vs PBC Comparison - selective uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0932	5.34	15.29
	PBC	A	0.0320	1.84	6.78
	FBL	B	0.0908	5.20	35.39
	PBC	B	0.0267	1.53	6.63
Bang-Bang	FBL	A	0.0924	5.29	16.91
	PBC	A	0.0351	2.01	7.05
	FBL	B	0.0901	5.16	32.77
	PBC	B	0.0271	1.55	6.89
Bang-Coast-Bang	FBL	A	0.0927	5.31	16.70
	PBC	A	0.0350	2.00	7.08
	FBL	B	0.0901	5.16	32.75
	PBC	B	0.0259	1.48	6.90

4.2 Uncertainty Level: EXTREME

4.2.1 Matched Initial Conditions, Optimized Gains

Table 23: FBL vs PBC Comparison - uniform uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0060	0.34	2.88
	PBC	A	0.0032	0.18	1.32
	FBL	B	0.0095	0.55	18.57
	PBC	B	0.0088	0.50	5.48
Bang-Bang	FBL	A	0.0058	0.33	2.72
	PBC	A	0.0030	0.17	1.38
	FBL	B	0.0093	0.54	20.56
	PBC	B	0.0085	0.49	5.44
Bang-Coast-Bang	FBL	A	0.0058	0.33	2.77
	PBC	A	0.0030	0.17	1.38
	FBL	B	0.0091	0.52	20.53
	PBC	B	0.0084	0.48	5.45

4.2.2 Matched Initial Conditions, Unoptimized Gains

Table 24: FBL vs PBC Comparison - uniform uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.1187	6.80	30.65
	PBC	A	0.0086	0.49	2.76
	FBL	B	0.1168	6.69	80.51
	PBC	B	0.0126	0.72	10.12
Bang-Bang	FBL	A	0.1186	6.79	30.99
	PBC	A	0.0084	0.48	2.84
	FBL	B	0.1161	6.65	82.66
	PBC	B	0.0125	0.72	10.38
Bang-Coast-Bang	FBL	A	0.1176	6.74	30.86
	PBC	A	0.0084	0.48	2.84
	FBL	B	0.1143	6.55	81.74
	PBC	B	0.0124	0.71	10.42

4.2.3 Mismatched Initial Conditions, Optimized Gains

Table 25: FBL vs PBC Comparison - uniform uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0265	1.52	2.88
	PBC	A	0.0314	1.80	1.32
	FBL	B	0.0227	1.30	18.60
	PBC	B	0.0340	1.95	5.48
Bang-Bang	FBL	A	0.0268	1.54	2.72
	PBC	A	0.0347	1.99	1.38
	FBL	B	0.0231	1.32	20.57
	PBC	B	0.0351	2.01	5.44
Bang-Coast-Bang	FBL	A	0.0267	1.53	2.77
	PBC	A	0.0343	1.97	1.38
	FBL	B	0.0230	1.32	20.53
	PBC	B	0.0353	2.02	5.45

4.2.4 Mismatched Initial Conditions, Unoptimized Gains

Table 26: FBL vs PBC Comparison - uniform uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.1184	6.78	34.02
	PBC	A	0.0327	1.87	2.76
	FBL	B	0.1157	6.63	64.76
	PBC	B	0.0292	1.67	10.12
Bang-Bang	FBL	A	0.1180	6.76	34.41
	PBC	A	0.0355	2.03	2.84
	FBL	B	0.1153	6.61	66.10
	PBC	B	0.0304	1.74	10.38
Bang-Coast-Bang	FBL	A	0.1170	6.71	34.27
	PBC	A	0.0354	2.03	2.84
	FBL	B	0.1143	6.55	66.40
	PBC	B	0.0304	1.74	10.42

5 Extreme Cases Analysis

5.1 Best Performance

Configuration:

- Controller: PBC
- Uncertainty: extreme
- Initial condition: matched
- Gains: optimized
- Trajectory: bang_coast_bang (label A)

Metrics:

- EE max error: 1.41 mm
- EE RMS error: 1.12 mm
- Joint max error: 0.34 deg
- Joint RMS error: 0.17 deg

5.2 Worst Performance

Configuration:

- Controller: FBL
- Uncertainty: extreme
- Initial condition: matched
- Gains: unoptimized
- Trajectory: bang_bang (label B)

Metrics:

- EE max error: **83.23** mm
- EE RMS error: 50.97 mm
- Joint max error: 17.23 deg
- Joint RMS error: 6.65 deg

6 Summary Comparison Table

This section presents a condensed head-to-head comparison using End-Effector Max Error [mm]. The better value is **Green**.

Table 27: Head-to-head controller comparison (lower is better)

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
BB /extreme/Match-Opt (A)	3.92	1.42	2.50
BB /extreme/Match-Opt (B)	20.56	6.61	13.95
BCB /extreme/Match-Opt (A)	3.88	1.41	2.48
BCB /extreme/Match-Opt (B)	20.53	6.60	13.93
Quintic /extreme/Match-Opt (A)	4.07	1.42	2.65
Quintic /extreme/Match-Opt (B)	20.06	6.56	13.50
BB /extreme/Match-Unopt (A)	39.27	2.88	36.39
BB /extreme/Match-Unopt (B)	83.23	14.60	68.63
BCB /extreme/Match-Unopt (A)	39.23	2.88	36.35
BCB /extreme/Match-Unopt (B)	82.70	14.55	68.15
Quintic /extreme/Match-Unopt (A)	39.80	2.89	36.92
Quintic /extreme/Match-Unopt (B)	81.15	14.56	66.59
BB /extreme/Mismatch-Opt (A)	20.84	20.84	0.00
BB /extreme/Mismatch-Opt (B)	20.84	20.84	0.00
BCB /extreme/Mismatch-Opt (A)	20.84	20.84	0.00
BCB /extreme/Mismatch-Opt (B)	20.84	20.84	0.00
Quintic /extreme/Mismatch-Opt (A)	20.84	20.84	0.00

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
Quintic /extreme/Mismatch-Opt (B)	20.84	20.84	0.00
BB /extreme/Mismatch-Unopt (A)	39.07	20.84	18.22
BB /extreme/Mismatch-Unopt (B)	81.44	20.84	60.60
BCB /extreme/Mismatch-Unopt (A)	39.05	20.84	18.21
BCB /extreme/Mismatch-Unopt (B)	81.17	20.84	60.32
Quintic /extreme/Mismatch-Unopt (A)	39.52	20.84	18.67
Quintic /extreme/Mismatch-Unopt (B)	82.13	20.84	61.28
BB /high/Match-Opt (A)	4.07	3.10	0.97
BB /high/Match-Opt (B)	11.46	4.11	7.36
BCB /high/Match-Opt (A)	4.07	3.09	0.98
BCB /high/Match-Opt (B)	11.40	4.09	7.31
Quintic /high/Match-Opt (A)	4.08	3.08	1.00
Quintic /high/Match-Opt (B)	11.45	4.09	7.36
BB /high/Match-Unopt (A)	21.04	7.31	13.74
BB /high/Match-Unopt (B)	48.51	7.89	40.61
BCB /high/Match-Unopt (A)	21.02	7.30	13.72
BCB /high/Match-Unopt (B)	48.56	7.85	40.71
Quintic /high/Match-Unopt (A)	20.93	7.31	13.62
Quintic /high/Match-Unopt (B)	48.28	7.87	40.41
BB /high/Mismatch-Opt (A)	20.84	20.84	0.00
BB /high/Mismatch-Opt (B)	20.84	20.84	0.00
BCB /high/Mismatch-Opt (A)	20.84	20.84	0.00
BCB /high/Mismatch-Opt (B)	20.84	20.84	0.00
Quintic /high/Mismatch-Opt (A)	20.84	20.84	0.00
Quintic /high/Mismatch-Opt (B)	20.84	20.84	0.00
BB /high/Mismatch-Unopt (A)	20.84	20.84	0.00
BB /high/Mismatch-Unopt (B)	48.51	20.84	27.67
BCB /high/Mismatch-Unopt (A)	20.84	20.84	0.00

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
BCB /high/Mismatch-Unopt (B)	48.56	20.84	27.72
Quintic /high/Mismatch-Unopt (A)	20.99	20.84	0.15
Quintic /high/Mismatch-Unopt (B)	48.26	20.84	27.42

6.1 Overall Winner Statistics

- FBL wins: 14 (29.2%)
- PBC wins: 34 (70.8%)
- Total comparisons: 48